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Batch 15

Algorithm

1. Start.
2. Read the decimal number `n`.
3. Call `decimalToBinary(n)`.
4. In the recursive function:
   - If `n == 0`, return.
   - Call `decimalToBinary(n / 2)`.
   - Print `n % 2`.
5. Stop.

Test Case Table

Test Case	Decimal Input	Expected Binary Output
1	0	0
2	1	1
3	5	101
4	10	1010
5	25	11001
6	100	1100100

C Program:

```
#include <stdio.h>

void decimalToBinary(int n)
{
    if (n == 0)
        return;
```

```
    decimalToBinary(n / 2);
    printf("%d", n % 2);
}

int main()
{
    int n;

    printf("Enter a decimal number: ");
    scanf("%d", &n);

    printf("Binary representation: ");

    if (n == 0)
        printf("0");
    else
        decimalToBinary(n);

    printf("\n");

    return 0;
}
```

Execution

